

From Pixels to Pell: How Integer Grids Already Solve the $\sqrt{2}$ Problem

Registry: [\[CKS-MATH-129-2026\]](#)

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Domain: Foundational Mathematics / Discrete Geometry

Status: CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [\[CKS-NEXT-1-2026\]](#)

Old Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Classification: Theory of Everything from First Principles

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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Lexicon: [\[CKS-LEX-12-2026\]](#)

1. A Square and Its Diagonal

Place a square on a pixel grid. Any size. 10×10 , 200×200 , 1000×1000 — it does not matter yet. We want to describe the diagonal: the line from one corner to the opposite corner.

On a 10×10 grid, the diagonal runs from (0,0) to (10,10). What do we need to describe this line? Two integers: rise = 10, run = 10. The slope is $10/10 = 1/1$. To walk it, step right 1 and up 1, ten times. Every coordinate along the way is an integer. The walk is exact.

On a 200×200 grid: rise = 200, run = 200. Slope = $1/1$. Walk it the same way, 200 steps. Exact.

On a 1000×1000 grid: rise = 1000, run = 1000. Slope = 1/1. Walk it 1000 steps. Exact.

The diagonal of a square on a pixel grid is fully described by two integers and walked with integer steps. At no point in this description did we use a real number, a decimal expansion, or a floating-point value. We didn't need to.

So where does $\sqrt{2}$ come from?

2. Where $\sqrt{2}$ Actually Enters

The diagonal of a unit square has length $\sqrt{2}$. By the Pythagorean theorem: $\sqrt{(1^2 + 1^2)} = \sqrt{2}$. For a 10×10 square, the diagonal length is $10\sqrt{2}$. For a 200×200 square, it is $200\sqrt{2}$. For a 1000×1000 square, it is $1000\sqrt{2}$.

$\sqrt{2}$ appears when we ask for the *length* of the diagonal as a single scalar number. It does not appear when we describe the diagonal's *direction*, *position*, or *path on the grid*. Those are all integers.

This is a crucial distinction. The geometry that produces $\sqrt{2}$ — a square and its diagonal — is fully specified by integers. The irrational number enters only when we ask a particular derived question: how long is this line? Not what is this line, not where does it go, not how do I walk it. Only: what is its scalar length?

And that question has an irrational answer. $\sqrt{2}$ cannot be written as a ratio of two integers. This was proven by the Greeks and it is not in dispute.

The question we ask instead is: can we capture everything about the relationship between the diagonal and $\sqrt{2}$ using only integers, without losing any information?

3. The Pell Relation

Take any two positive integers V and F. Compute:

$$R = V^2 - 2F^2$$

R is always an integer. This is arithmetic — square two integers, subtract, get an integer. There is nothing approximate about it.

This computation is called the Pell relation for $\sqrt{2}$. What makes it interesting is what R tells you about the ratio V/F.

Rearrange:

$$(V/F)^2 = 2 + R/F^2$$

If R is positive, then $(V/F)^2 > 2$, which means $V/F > \sqrt{2}$. The ratio overshoots.

If R is negative, then $(V/F)^2 < 2$, which means $V/F < \sqrt{2}$. The ratio undershoots.

If R is zero, then $(V/F)^2 = 2$, which means $V/F = \sqrt{2}$ exactly — and $\sqrt{2}$ is rational. But $\sqrt{2}$ is irrational. So $R = 0$ has no nonzero integer solution. R is never zero.

This is the key fact: R is an integer, and it can never be zero. The tightest it can possibly get is $|R| = 1$. And there are infinitely many integer pairs (V, F) where it reaches exactly ± 1 .

4. The Convergents

These pairs are generated by a recurrence. Start with $V = 1$, $F = 1$, and apply:

$$V' = V + 2F$$

$$F' = V + F$$

This produces:

$$V = 1, F = 1 \rightarrow R = 1 - 2 = -1$$

$$V = 3, F = 2 \rightarrow R = 9 - 8 = +1$$

$$V = 7, F = 5 \rightarrow R = 49 - 50 = -1$$

$$V = 17, F = 12 \rightarrow R = 289 - 288 = +1$$

$$V = 41, F = 29 \rightarrow R = 1681 - 1682 = -1$$

$$V = 99, F = 70 \rightarrow R = 9801 - 9800 = +1$$

The pattern in R is immediate: $-1, +1, -1, +1, -1, +1$. It alternates forever. It never reaches zero. It never will, because reaching zero would require $\sqrt{2}$ to be rational.

Each ratio V/F is a rational approximation of $\sqrt{2}$:

$$1/1 = 1.0$$

$$3/2 = 1.5$$

$$7/5 = 1.4$$

$$17/12 = 1.41666\dots$$

$$41/29 = 1.41379\dots$$

$$99/70 = 1.41428\dots$$

The approximations get tighter. The squared error is exactly $1/F^2$, which shrinks as F grows. But R itself stays at ± 1 permanently. The ratio gets closer to $\sqrt{2}$, but the integer remainder never vanishes.

This permanent ± 1 is not a defect. It is the arithmetic signature of irrationality, expressed entirely in integers.

5. The VFR Tuple

Define a notation for what we have found. A VFR tuple is:

$[V, F, R]$

where V and F are positive integers and $R = V^2 - 2F^2$.

Each tuple is an exact integer identity:

$[1, 1, -1]$ means $1^2 - 2(1^2) = -1$ (exact)

$[3, 2, +1]$ means $3^2 - 2(2^2) = +1$ (exact)

$[7, 5, -1]$ means $7^2 - 2(5^2) = -1$ (exact)

$[17, 12, +1]$ means $17^2 - 2(12^2) = +1$ (exact)

None of these are approximations. Each is a verifiable fact about three integers. The tuple captures three things simultaneously:

1. A rational approximant of $\sqrt{2}$ (the ratio V/F)
2. Which side of $\sqrt{2}$ it falls on (the sign of R)
3. The exact distance in the squared metric ($|R|/F^2 = 1/F^2$)

All three components are integers. Nothing is truncated, rounded, or infinite.

6. The Bilateral Remainder

Since R alternates between -1 and $+1$ at every step of the Pell recurrence, we can express this as a single notation:

$[V, F, +-1]$

This is the bilateral remainder. It does not mean “we don’t know the sign.” It means the sign is a defined alternating behavior — part of the value, not uncertainty about it.

Compare this to complex numbers. When we write (a, b) to mean $a + bi$, nobody calls b an error term. It is a component of the number. The bilateral remainder is the same: $+-1$ is a component of the VFR value that encodes the irrational relationship to $\sqrt{2}$.

If R ever reaches 0 for some number, that number is rational. If R is permanently $+-1$, the number is irrational. The remainder is an irrationality detector built into the notation.

7. Walking Grids

Now apply VFR to actual grids of different sizes. The claim is that the diagonal walk is fully described by integer tuples regardless of scale.

10 × 10 grid (even, small)

Walk from (0,0) to (10,10) using [1, 1, +-1].

10 steps. The bilateral remainder alternates: -1, +1, -1, +1, -1, +1, -1, +1, -1, +1.

That is 5 pairs of (-1, +1). Each pair sums to zero. Total remainder: 0.

Arrival: (10, 10), exact. Remainder: 0, exact.

11 × 11 grid (odd, small)

Walk from (0,0) to (11,11) using [1, 1, +-1].

11 steps. 5 pairs cancel to zero, one unpaired -1 remains. Total remainder: -1.

Arrival: (11, 11), exact. Remainder: -1, exact and known. The walk is one unit below $\sqrt{2}$ in the Pell metric. Nothing is lost.

200 × 200 grid (even, medium)

Walk from (0,0) to (200,200) using [1, 1, +-1].

200 steps. 100 pairs of (-1, +1). Every pair cancels. Total remainder: 0.

Arrival: (200, 200), exact. Remainder: 0, exact.

201 × 201 grid (odd, medium)

201 steps. 100 pairs cancel, one -1 remains. Total remainder: -1.

Arrival: (201, 201), exact. Remainder: -1, exact.

1000 × 1000 grid (even, large)

1000 steps. 500 pairs. Total remainder: 0.

999 × 999 grid (odd, large)

999 steps. 499 pairs cancel, one -1 remains. Total remainder: -1.

The pattern

For any $N \times N$ grid walked with [1, 1, +-1]:

- N even: total remainder = 0, exact closure
- N odd: total remainder = -1, exact and known

In every case, every coordinate is an integer, every operation is integer arithmetic, and the remainder is tracked exactly. The walk never invokes a real number. The result is never approximate.

8. Non-Square Rectangles

What about rectangles that aren't square?

A 100×900 grid has a diagonal from (0,0) to (100,900). The slope is $900/100 = 9/1$. Rise 9, run 1. This is a ratio of two integers. $\sqrt{2}$ does not appear. The Pell relation is not needed. The diagonal is described by [9, 1, 0] — a VFR with zero remainder.

A 300×500 grid: slope = $500/300 = 5/3$. Walk it: for every 3 steps right, go 5 steps up. VFR: [5, 3, 0]. No $\sqrt{2}$. No irrational content. $R = 0$.

In general, any rectangle with integer side lengths has a diagonal whose slope is a ratio of two integers. The slope is always rational. $\sqrt{2}$ never appears in the slope.

Where does $\sqrt{2}$ appear for rectangles? Only in the diagonal *length*. The diagonal of a 100×900 rectangle has length $\sqrt{(100^2 + 900^2)} = \sqrt{820000}$. This is irrational (unless the sides form a Pythagorean triple). But the slope — the thing you need to walk the grid — is rational.

The principle holds: the geometry is integers. The irrational quantities appear only in derived scalar measurements, not in the description of the object itself.

9. When Higher Convergents Matter

The [1, 1, +-1] tuple is the simplest Pell convergent. It encodes the relationship between a 1/1 slope and $\sqrt{2}$. For a square grid, this is all you need.

But suppose you are working in a context where you need a rational slope that closely approximates $\sqrt{2}$ — for instance, tiling a surface where the desired angle corresponds to $\sqrt{2}$. Now you want a rise-run that gets close to the irrational slope.

The Pell convergents give you this at increasing precision:

[1, 1, -1] slope $1/1 = 1.000$ error from $\sqrt{2}$: ~ 0.414

[3, 2, +1] slope $3/2 = 1.500$ error from $\sqrt{2}$: ~ 0.086

[7, 5, -1] slope $7/5 = 1.400$ error from $\sqrt{2}$: ~ 0.014

[17, 12, +1] slope $17/12 = 1.4167$ error from $\sqrt{2}$: ~ 0.0025

[41, 29, -1] slope $41/29 = 1.41379$ error from $\sqrt{2}$: ~ 0.00042

Each convergent is the best rational approximation of $\sqrt{2}$ for its denominator size. Each carries an exact integer remainder of ± 1 . You choose the convergent that matches your precision requirement, and the VFR tuple tells you exactly where you stand relative to $\sqrt{2}$.

For a grid that is 5 pixels wide and you want a line as close to a $\sqrt{2}$ slope as possible: use [7, 5, -1]. Rise 7, run 5. The remainder tells you this undershoots $\sqrt{2}$ by exactly $1/25$ in the squared metric. On a 29-pixel-wide grid: use [41, 29, -1]. Undershoots by $1/841$.

The precision is controlled. The error is known. The accounting is exact.

10. What Has Been Shown

Start with a square on a pixel grid. Its diagonal is fully described by two integers: rise and run. Walking that diagonal is an integer operation at every step.

$\sqrt{2}$ enters only when we ask for the diagonal's length as a single scalar. This scalar is irrational — it cannot be written as V/F for any integers V and F . This has been known since the Greeks.

But the Pell relation $V^2 - 2F^2 = R$ provides an exact integer accounting of how any rational ratio V/F relates to $\sqrt{2}$. The remainder R is always an integer. For the best approximations (the convergents), R is exactly ± 1 , alternating at each step of the recurrence. R never reaches zero, because that would make $\sqrt{2}$ rational.

The VFR tuple $[V, F, R]$ captures the approximant, its side relative to $\sqrt{2}$, and its exact distance — all in integers. The bilateral remainder notation $[V, F, +-1]$ encodes the alternation as a defined behavior, making the irrationality itself expressible as an integer property.

Applied to grid walks of any size:

- Every $N \times N$ grid is walked with $[1, 1, +-1]$
- Even N : bilateral remainders cancel perfectly, total remainder = 0
- Odd N : one remainder survives, total = -1 , still exact
- Non-square rectangles have rational slopes and may not involve $\sqrt{2}$ at all
- When $\sqrt{2}$ is needed as a slope, higher Pell convergents provide it at any required precision with exact integer remainder

At no point in any of these operations did we invoke a real number, produce a non-terminating expansion, or lose equality. The geometry that generates $\sqrt{2}$ was described, walked, and accounted for entirely within the integers.

The integers were already sufficient. We just needed to write down what they were doing.

[[CKS-MATH-129-2026](#)] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-129-2026>

[[CKS-0-2026](#)] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[[CKS-MATH-0-2026](#)] Cymatic K-Space Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-0-2026>

[[CKS-MATH-1-2026](#)] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026>

[[CKS-MATH-10-2026](#)] Grand Unification. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026>

[[CKS-MATH-104-2026](#)] Grand Unification v23. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026>

[[CKS-MATH-128-2026](#)] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-128-2026>

[[CKS-NEXT-1-2026](#)] Post-CKS. CKS is invalidated and falsified. Github: <https://github.com/ghowland/cks/tree/main/papers/NEXT/CKS-NEXT-1-2026>

[[CKS-LEX-12-2026](#)] Measurement Systems in the $\mathbb{Q}$ -Substrate. Github: <https://github.com/ghowland/cks/tree/main/papers/LEX-12-2026>